Triangles

Core Concepts: Triangles

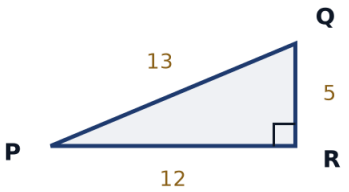
1 In $\triangle ABC$ studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 81°

b. 91°

c. 73°

d. 126°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm

c. 24 cm

d. 32 cm

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 13

c. 11

d. 15

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 74°

b. 84°

c. 66°

d. 126°

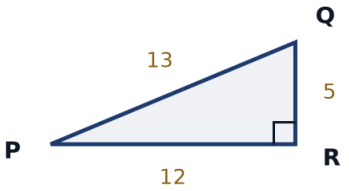
5 A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm

c. 32 cm

d. 72 cm



6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 11

b. 17

c. 15

d. 19

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 67°

b. 77°

c. 59°

d. 126°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

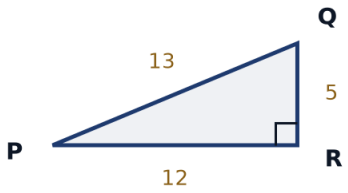
9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 15

b. 21

c. 19

d. 23



Exam-Style Problem Solving & Modeling

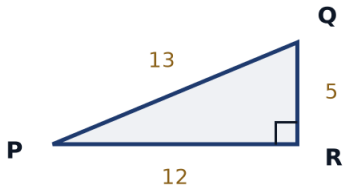
10 In $\triangle ABC$ studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 60°

b. 70°

c. 52°

d. 126°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm

c. 48 cm

d. 200 cm

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 19

b. 25

c. 23

d. 27

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